



Original Article

Maternal presence facilitates plasticity in offspring behavior: insights into the evolution of parental care

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Fundamental to the definition of parental care is that care confers benefits to the offspring. However, the mechanisms resulting in these benefits remain poorly understood, particularly in species where postnatal care is not obligatory. Here, we address this shortcoming using a lizard, *Liopholis whitii*, in which family life is facultative and relatively simple—extending to prolonged associations between parents and offspring within the parental territory. Using a split-clutch design, we housed offspring either with their mother or alone during the first 8 weeks of postnatal life and examined whether maternal presence affected 1) the expression of key functional behaviors and 2) learning ability in a biologically relevant antipredatory task. We found that offspring housed with their mothers expressed heightened levels of activity, boldness, and exploration compared with offspring who were housed alone. Furthermore, we show that associating with mothers during early postnatal periods led to improved offspring performance in the antipredation learning task. Together these results suggest that even relatively simple forms of enhanced parent-offspring association can have significant impacts on offspring traits. We argue that such effects may help refine and stabilize parent-offspring associations early in their evolution, potentially setting the stage for the elaboration of both parent and offspring behaviors.

Key words: development, learning, parental care, sociality, reptile.

INTRODUCTION

Parental care is taxonomically and functionally diverse. At a functional level, it ranges from the simple provisioning of gametes to extended investment in offspring long after nutritional dependence (Royle et al., 2012a). At a taxonomic level, parental care is exhibited across a range of organisms, with mammals, birds, fish, and insects representing the groups with the most complex forms of care (Clutton-Brock 1991; Reynolds et al. 2002). Even within these groups, parental care can vary considerably from being exhibited by only females (as in most mammal species), only males (as in many fish species), or by both sexes (as in many bird species) (Clutton-Brock 1991; Royle et al., 2012a; see Kokko and Jennions (2008) and Klug et al. (2013) for theoretical models explaining this). Parental care is also a fundamental characteristic of broader family life, and as a consequence provides a nexus point for the adaptive radiation of kin relationships and sociality (Queller 1994; Field and Brace 2004; Halliwell et al. 2017a). As a result, there is significant

interest in understanding the factors responsible for the emergence and diversification of parental care.

For vertebrates, a key focus has been on the emergence and diversification of postbirth parental care. Such care emerges primarily (but not always; Clutton-Brock 1991; Royle et al. 2012a) when parents regularly encounter and associate with their offspring (Lion and van Baalen 2007; Halliwell et al. 2017a) and when the benefits of social interactions with parents for offspring are not outweighed by the costs to the parents (Clutton-Brock 1991). Therefore, conditions that promote an increased level of association between parents and their offspring and mediate the costs and benefits of those associations will be central to facilitating the early emergence of care (Lion and van Baalen 2007; Davis et al. 2011; Royle et al. 2012b; Klug and Bonsall 2014; Halliwell et al. 2017a). Where these conditions are recurrent, parent-offspring associations can create a novel social environment from which a range of behavioral, cognitive, physiological, and morphological traits, in both parents and offspring, can be refined and co-opted to function in parental care (Carere and Locurto 2011; Gardner and Smiseth 2011; Uller 2012; Royle et al. 2012b; Curley and Champagne 2016).

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Central to understanding the early evolution of parental care is to thus 1) identify the factors that promote the initial increase in associations between parents and offspring, and 2) understand how the emergence of simple parent-offspring associations provide benefits to offspring that may help stabilize those associations. Phenotypic plasticity will be of particular importance in both these contexts. First, the initiation of parent-offspring associations will be aided by plastic responses that facilitate changes in how parents and offspring interact with one another (e.g., social plasticity; Royle et al. 2014). For example, delayed habitat availability can mediate plastic dispersal behavior and facilitate a greater level of interaction between parents and offspring (e.g., Botterill-James et al. (2016), Kingma et al. (2016), and Halliwell et al. (2017b)). Second, phenotypic plasticity will be fundamental for the stabilization of these associations if the increase in parent-offspring association facilitates the plastic expression of, or the development of, offspring phenotypes that provide benefits to offspring (Royle et al. 2012b), potentially at a minimal cost to the parent (see Botterill-James et al. (2016)). Such effects can manifest themselves in both the short- and long-term. In the short-term, the presence of a parent may have activational effects on the offspring, allowing them to display traits that provide greater access to resources (see Webster and Ward (2011)). In the long-term, the presence of a parent may have organizational effects on offspring traits, by providing a novel environmental context that affects trait development (e.g., offspring can devote more time to learning foraging and antipredatory behaviors; Hoppitt et al. 2008). Indeed, work on obligate social species has shown that parent-offspring associations during early life can have substantial effects on the development of key traits (e.g., behavior and cognition), with important implications for fitness (reviewed in Clutton-Brock (1991) and Curley and Champagne (2016)). What role such effects have during the early evolution of parental care is currently unknown but they have the potential to help refine and stabilize parent-offspring associations.

Here we addressed this question by examining how simple associations between parents and offspring mediate offspring behavior and development and test the extent to which this provides benefits to offspring during the early stages of life. To achieve this, we used a family living lizard, *Liopholis whitii* (previously *Egernia whitii*) as our model system. Lizards are excellent model systems in this context. First, family life is facultative and relatively simple compared with other systems, typically extending to parents tolerating offspring within their home ranges (e.g., an increase in parent–offspring associations; [While et al. 2014](#)). Second, these parent–offspring associations are relatively widespread phylogenetically having evolved multiple times independently from an ancestral state of no care ([Halliwell et al. 2007a](#)). Thus, lizards have the potential to provide fundamental insights into the factors that may have facilitated the early emergence of parental care. The best-studied group in this context is the *Egernia* group, a group of family-living lizards. Family life in *Egernia* is characterized by long-term, stable male–female pair bonds and prolonged parent–offspring associations ([Chapple 2003](#); [Whiting and While 2017](#)). Furthermore, there is considerable variation in the duration of these parent–offspring associations, from species that exhibit no associations between parents and offspring to species which exhibit extended, multiyear, associations ([Chapple 2003](#); [Whiting and While 2017](#)). In *L. whitii*, females typically give birth to clutches of up to four offspring, most of these offspring disperse immediately after birth while a single offspring remains in the parental territory (usually for a single year but sometime for up to several years; [Chapple 2003](#); [While et al. 2009a](#)). This prolonged

period of association with parents is associated with significant benefits for the offspring in terms of decreased aggression from conspecifics (O'Connor and Shine 2004; Sinn et al. 2008) and increased growth rates (Botterill-James et al. 2016), whereas there appears to be limited costs to the parents (Botterill-James et al. 2016).

Combined, the above makes *L. whittii* an excellent system to test sociodevelopmental questions regarding the mechanisms that mediate the consequences of parent–offspring associations for both the parents and offspring. To exploit this, we manipulated the presence/absence of parent–offspring association by raising offspring with and without their mother over the first 8 weeks of life (a crucial period of development; Botterill-James et al. 2016; Munch et al. 2018a). We then examined the extent to which parent–offspring associations provided benefits to offspring in terms of increased offspring growth and the extent to which these benefits were underpinned by the context-dependent expression of ecologically relevant behavioral (i.e., boldness, activity, and exploration) and cognitive traits. These traits have previously been shown to increase resource availability (Biro et al. 2006; Stamps 2007; Boon et al. 2007) but also increase exposure to predation (Riechert 2000). As a key function of parent–offspring associations in this system is reduced predation risk, via reduced risk of conspecific aggression (O'Connor and Shine 2004; Sinn et al. 2008), we predicted that offspring who associate with their parents would benefit from the protected period of life. Benefits would arise through the enhanced expression behavioral traits and improved learning ability and this would ultimately translate into increased fitness (i.e., faster growth).

METHODS

Study species

Liopholis whitii is a medium-sized (up to 100-mm snout–vent length [SVL]), viviparous lizard that is widely distributed throughout a broad altitudinal (0–1600 m) and habitat (coastal heath, grasslands, and forests) range in south-eastern Australia, including Tasmania. *Liopholis whitii* are generally found in close association with rock outcrops where they excavate deep and complex burrows under or at the base of rock and shrub vegetation, which are used as permanent shelters and retreat sites (Chapple 2003; Wilson and Swan 2003). Males and females are monomorphic and reach sexual maturity at approximately 3 years. Reproduction occurs annually, with mating during the austral spring (September–October) and gestation spanning 3–4 months (While et al. 2009a). Parturition occurs in the Austral summer (January–February) with clutches comprising one to four offspring born asynchronously, usually over several days (While et al. 2007).

Experimental protocol

We caught 22 gravid females at the end of gestation (mid-January) from populations on the east coast of Tasmania (42°57'S, 147°88'E and 42°22'S, 147°77'E). Females were transported to the University of Tasmania where they were measured for weight (± 1 mg) and length (SVL and total length; ± 0.5 mm), and toe-clipped to allow for unique identification. Females were housed individually in opaque terraria [57 (L) $\times$ 38 (W) $\times$ 32 (H) cm], provided with cat litter as a substrate (~ 5 -cm thick), a rock as a basking rock and a refuge (Terracotta saucer). Basking lights (25 W) and overhead UV lighting set to 0800–1800 h night/day cycle provided thermoregulatory opportunities. Water and food (*Tenebrio* larvae and fruit puree mixed with protein powder) were provided ad libitum.

We checked terraria twice daily for offspring. At birth, we recorded offspring morphometric traits (weight [± 1 mg] and length [SVL and total length; ± 0.5 mm]) and an offspring's birth order in the clutch was recorded. Offspring were then given a unique toe-clip for later identification. We were unable to identify sex of offspring as juvenile female *L. whitii* retain their hemipenes until well after birth (Chapple 2005).

Social development experiment

A total of 52 offspring were born to the 22 females held in the laboratory (average clutch size = 2.36 ± 0.14 ; average birth spread = 3.2 days ± 0.46). We randomly assigned the first and second born offspring from each clutch into 1 of 2 social treatments: 1) offspring were housed together with their mother ($n = 22$, "maternal presence") or 2) housed alone ($n = 20$, "maternal absence"). These treatments reflect the variation in parent-offspring association that is observed in the natural populations of *L. whitii* where often only a single offspring from a clutch will remain within the maternal territory while the other offspring disperse (While et al. 2009b). Two females gave birth to a single offspring, for these females we allocated both offspring into the "maternal" treatment. We ensured that including these 2 offspring in our analysis did not influence our result (see Supplementary Tables S1–S3).

All lizards (mothers and offspring) were housed for the duration of the social treatment (8 weeks) in small outdoor enclosures (1-m diameter) at the University of Tasmania. Enclosures were covered with bird netting to prevent predation, and each enclosure was supplied with sand as substrate, a hollow concrete brick (Bessa block) for basking and shelter, as well as a 30- $\times$ 15-cm piece of corrugated iron for shelter. Water was provided ad libitum and food (a handful of *Tenebrio* larvae) was provided 3 times weekly early in the morning before the lizards were out basking. Five offspring died during the experiment, resulting in a final sample size of 38 offspring ($n = 19$ "maternal presence"; $n = 19$ "maternal absence").

Behavioral trials

During the social development experiment, we examined whether the social context that offspring experienced affected the expression of 3 key behaviors often correlated with life history traits such as growth, survival, and reproduction (Stamps 2007; Réale et al. 2009).

- 1) **Boldness:** Boldness was scored as the latency in time (s) until offspring emerged after a scare. Specifically, offspring were scared underneath their shelter by the experimenter simulating a predatory attack. The amount of time it took for offspring to emerge was recorded as a measure of boldness with lower values indicating bolder behavior. If offspring did not emerge during the 30-min period, they were assigned a maximum emergence time of 1800 s.
- 2) **Activity:** Activity was scored as the length of time (s) spent moving in the familiar environment.
- 3) **Exploration:** Exploration was scored as the length of time (s) spent moving in a novel environment.

Offspring in the maternal association treatment were tested in the presence of their mother. Trials were conducted in the offspring's enclosure during their active period between 1000 and 1300 h (While GM, personal observation). As trials were conducted in semi-natural conditions, we were unable to control for environmental

conditions that could have influenced the expression of the behavioral traits (e.g., temperature). However, trials were only conducted on sunny days, and offspring from the 2 treatments were all assessed on the same day and randomly throughout the experimental period so that any effects of temperature would be evenly spread across treatments. Trials for each behavior ran separately for 30 min and were filmed using GoPro Hero4 cameras (CA, USA). Boldness and activity trials were conducted 3 times over the 8-week period of the social developmental experiment, starting a week after the offspring were released into their enclosure, with approximately 10 days of rest between the trials (i.e., exploration: day 0; boldness: days 10, 30, and 50; and activity: days 20, 40, and 60). We included trial number in all analyses of these 2 behaviors to account for any effect of habitation over the 3 time periods and minimized the presence of the experimenter during this period. The exploration assay was conducted only once when the offspring were first released into the center of their enclosure, as this was the only time the environment was novel to the offspring (see Rodríguez-Prieto et al. (2011) for similar approach). All videos were scored by a single observer (A.R.).

Antipredatory learning task

Following completion of the social development experiment, offspring were returned to the laboratory to have their morphometric traits remeasured and to undergo an antipredation learning task. In the laboratory offspring were housed individually in opaque tubs [57 (L) $\times$ 38 (W) $\times$ 32 (H) cm], kept under a 25-W basking light set to a 0800–1800 h night/day cycle. Each tub contained cat litter as a substrate and a rock as a basking site and a refuge. Water was provided ad libitum and food was provided 3 times weekly.

We assayed offspring learning ability using a biologically relevant antipredatory task. In this task, offspring were given the option of 2 refuges (see below for the definition of "safe" and "unsafe") over 20 trials to determine the amount of trials it took for the offspring to learn the location of a "safe" refuge (see Holtzman et al. (1999), Paulissen (2008), Noble et al. (2012), and Carazo et al. (2014) for similar approaches). Offspring were able to spatially orient themselves using extra-maze positional cues (i.e., room features and placement of experimenter conducting the trials). Such cues (i.e., left or right placement of features) have previously been shown to expedite discriminatory learning in lizards (Day et al. 2003). This learning task was carried out in a common garden in the offspring's home enclosure, without the presence of the mother. This allowed us to examine the role that the presence of a mother had on the development of key functional traits (e.g., learning) as opposed to facultative behavioral response to the presence/absence of a mother (which we tested above).

At the beginning of each trial, we removed the basking rock, food, and water dishes, and gently chased the offspring into its shelter (placed in the center of the enclosure). We then added the 2 refuges (11-cm diameter Terracotta saucers with an opening on one side) at opposite ends of the enclosure. We removed the center shelter shielding the offspring and left the offspring to acclimate for 5 min before the antipredatory trial began (see below for how we dealt with instances where offspring were found inside a refuge at the end of the acclimation period). After this time, we simulated a predatory attack following a standardized protocol by chasing the offspring around its enclosure by gently tapping its pelvic girdle from behind with our index finger until it entered the "safe" refuge. If the offspring entered the "unsafe" refuge, we then lifted the refuge, tapped the offspring out, and immediately returned the refuge

in its original position and resumed chasing the offspring. We gave offspring a maximum of 10 min to complete the task. Once the offspring entered the “safe” refuge, we removed the “unsafe” refuge and added the central shelter, basking rock, and food and water dishes. We left the “safe” refuge inside the enclosure until the offspring reemerged from hiding, at which point we gently removed the refuge taking care not to startle the offspring. We scored the number of times an offspring chose the “unsafe” refuge before choosing the “safe” refuge (e.g., number of errors). If an offspring was found inside the “safe” refuge prior to the end of the acclimation period (109 times out of 760 observations [14.34%]), we considered it as having made a correct choice. We argue that this is a biologically sensible interpretation, as we expect offspring would learn the location of the safe “refuge,” along with the procedure, over time and take refuge; excluding these observations did not affect our results (Supplementary Table S4). If an offspring was found inside the “unsafe” refuge prior to the end of the acclimation period (247 times out of 760 observations [32.50%]), we considered it to have chosen incorrect and began the predatory attacks. If an offspring did not choose any refuge during the trial, we gently forced it into the “safe” refuge but scored it as having made an incorrect choice; again, excluding these observations did not affect our result (Supplementary Table S5).

We mixed cage substrate prior to each trial and washed the refuges with 80% ethanol immediately after each trial to control for offspring’s use of chemosensory cues. For all trials, we kept the position of both refuges constant throughout the task but randomized the location of which refuge was “safe” and “unsafe” between offspring. Offspring were assayed twice daily (morning and afternoon) over 10 consecutive days for each task ($n = 20$ trials) as previous studies have shown this to be an appropriate number of trials to estimate differences in learning ability (Noble et al. 2012). All trials were conducted blind to the offspring’s developmental treatment. Three experimenters conducted the task but were randomly allocated between offspring each trial.

At the completion of the learning task (early May), all lizards (mothers and offspring) were released back at their site of capture.

Statistical analysis

We first examined whether offspring differed in their expression of the 3 behavioral traits (activity, boldness, and exploration) depending on their social context using mixed-effects models. Exploration was log-transformed to make error distributions approximately normal and modeled using a linear mixed model (LMM). Due to zero-inflation of the count data for boldness and activity, we used generalized linear mixed model (GLMMs) with zero-adjusted negative binomial distributions. We included offspring treatment in all models together with SVL at birth as previous studies have shown that behavioral traits may covary with body size (Amiel et al. 2014). Maternal identity was fitted as a random effect. For behaviors where we had repeated observations (i.e., activity and boldness), we included trial number as a fixed effect to account for any effects of habituation over time and offspring identity as a random effect (Biro 2012). We assessed the adjusted repeatability of activity and boldness using the “rptR” package (Stoffel et al. 2017). Due to zero-inflation, we were unable to model repeatability in a single model. Instead we divided our data into 2 datasets. First, we analyzed whether or not offspring performed the behavior (i.e., 0 or 1) using a binomial/Bernoulli error distribution, whereas values equal to 1 or above were analyzed using a “Poisson” error distribution. In

both the binary and Poisson models, we included treatment as a fixed factor and offspring identity as a grouping factor.

Second, we examined whether offspring differed in their performance in the antipredatory learning task depending on whether they had been raised with or without their mother. This was achieved by modeling the number of errors made by offspring before choosing the “safe” refuge using a GLMM with a zero-adjusted negative binomial error distribution. We fitted all models with offspring social treatment, trial number, and an interaction between treatment and trial to test for differences in rates of learning. This approach explicitly allows us to incorporate both between- and within-individual variations in number of errors, improving power, whilst still estimating treatment-level changes (see Kar et al. (2017) for similar approaches; Riley et al. 2017). If offspring were learning, we predicted that the number of errors would decrease across trials. We included SVL at the end of the social developmental experiment (i.e., 8 weeks) as a fixed effect to account for differences in body size, a fitness-related trait that is argued to affect learning abilities in reptiles (Amiel et al. 2014). We also included which side the “safe” refuge was located to account for any lateralization bias (see Supplementary Material). Individual and maternal identity were included as random effects. Initially we included a random intercept and slope for trial number within each offspring to account for nonindependence of observations on the same offspring; however, this led to poor model convergence and was subsequently dropped from the analyses. Experimenter identity was initially included in the models as a fixed effect to account for any differences between the experimenters conducting the trials; this did not affect the result ($\chi^2 = 4.10$, $df = 2$, $P = 0.13$; Supplementary Table S6), and so, we present the reduced model without this variable.

We also tested whether offspring growth (i.e., change in body size [SVL] from birth to end of the social treatment) was affected by the social treatment using a LMM. We used SVL as this is the most widely used measure of body size in reptiles (offspring SVL and body mass were correlated; $R = 0.55$, $CI = 0.28–0.74$, $P < 0.001$). We fitted number of days between measurements as a covariate, and maternal identity as a random effect due to our split-clutch design.

All analyses were carried out in R version 3.3.2 (R Core Team 2016). LMMs for exploration and GLMM for growth were modeled with the “lme4” package (Bates et al. 2015). Zero-adjusted negative binomial GLMMs for exploration, boldness, and learning were modeled using the “glmmTMB” package (Brooks et al. 2017). For all LMMs, the significance of fixed effects is reported based on F -tests with degrees of freedom approximated by Kenward–Rodgers method. For GLMMs, the significance of fixed effects is reported based on likelihood ratio tests (LRT). All continuous variables (i.e., SVL and trial number) were z -transformed prior to analysis by mean centering and dividing by their standard deviation to improve model interpretation; this does not influence inferential statistics (Schielzeth 2010).

RESULTS

We found that maternal presence had a number of significant effects on offspring behavior both during and at the end of the experimental treatment. First, we found that the presence of a mother significantly influenced the expression of all 3 behavior traits. Offspring were significantly bolder (i.e., they emerged quicker after a scare; $\chi^2 = 7.14$, $df = 1$, $P < 0.01$), more active

(i.e., they spent more time moving; $\chi^2 = 41.56$, $df = 1$, $P < 0.001$), and more exploratory (i.e., they spent more time exploring a novel environment; $F_{1,19.43} = 5.52$, $P = 0.03$) when in the presence of their mother compared with offspring who were housed alone (Table 1; Figure 1). Within these treatments, however, we found no evidence that these behaviors were repeatable, with repeatability ranging from $R_{adj} = 0.01$ (binary, CI = 0–0.27, $P = 0.46$) and $R_{adj} = 0$ (Poisson, CI = 0–0.09, $P = 0.50$) for boldness and $R_{adj} = 0.01$ (binary, CI = 0–0.75, $P = 0.50$) and $R_{adj} = 0.12$ (Poisson, CI = 0–0.10, $P = 0.33$) for activity. The low repeatability could suggest two things, either that the social context explained the majority of the between-individual variation in our data or that these behavioral responses were largely random within treatments.

Offspring who associated with their mother during the first 8 weeks of life also performed better in the antipredator task at the end of the experiment compared with offspring who had been alone. Specifically, offspring that associated with their mothers made significantly less errors in choosing the correct refuge across time compared with offspring that were housed by themselves,

who slightly increased in the number of errors made over time (trial $\times$ treatment interaction: $\chi^2 = 4.92$, $df = 1$, $P = 0.03$; Table 2; Figure 2).

Finally, maternal presence had a negative effect on offspring growth, with offspring housed by themselves growing significantly more than offspring housed with their mother ($F_{1,19.99} = 15.35$, $P < 0.001$; Figure 3).

DISCUSSION

The social environment during the early period of life can mediate offspring phenotypic development in a number of ways, either through behavioral interactions and information acquisition with other individuals or via the mere presence of other individuals (Bester-Meredith and Marler 2001; Curley and Champagne 2016). Importantly, such effects have been suggested to have significant implications for the evolution of parental care (Royle et al. 2012b). Here we show that in a family-living lizard, *L. whiti*, where parental care extends to simple tolerance of offspring within the home range

Table 1
Estimates and 95% confidence intervals from GLMMs and LMMs examining the effect of social context (maternal absence vs. presence [MUM]) on offspring behavioral traits

	(a) Boldness			(b) Activity			(c) Exploration		
	Estimate	Lower	Upper	Estimate	Lower	Upper	Estimate	Lower	Upper
Intercept	5.75	5.26	6.24	2.40	2.08	2.72	3.51	3.09	3.94
SVL	0.07	−0.26	0.41	0.03	−0.15	0.21	0.10	−0.20	0.22
Treatment MUM	−0.93	−1.61	−0.26	1.72	1.31	2.13	0.61	0.11	1.11
Trial number	−0.38	−0.68	−0.08	0.10	−0.09	0.29	—	—	—

(a) Boldness (i.e., latency to emerge; $N_{obs} = 101$ from 38 offspring), (b) activity (i.e., time spent moving; $N_{obs} = 101$ from 38 offspring), and (c) exploration (i.e., time spent moving in a novel environment; $N_{obs} = 38$ from 38 offspring). Statistically significant values ($P < 0.05$) are highlighted in bold.

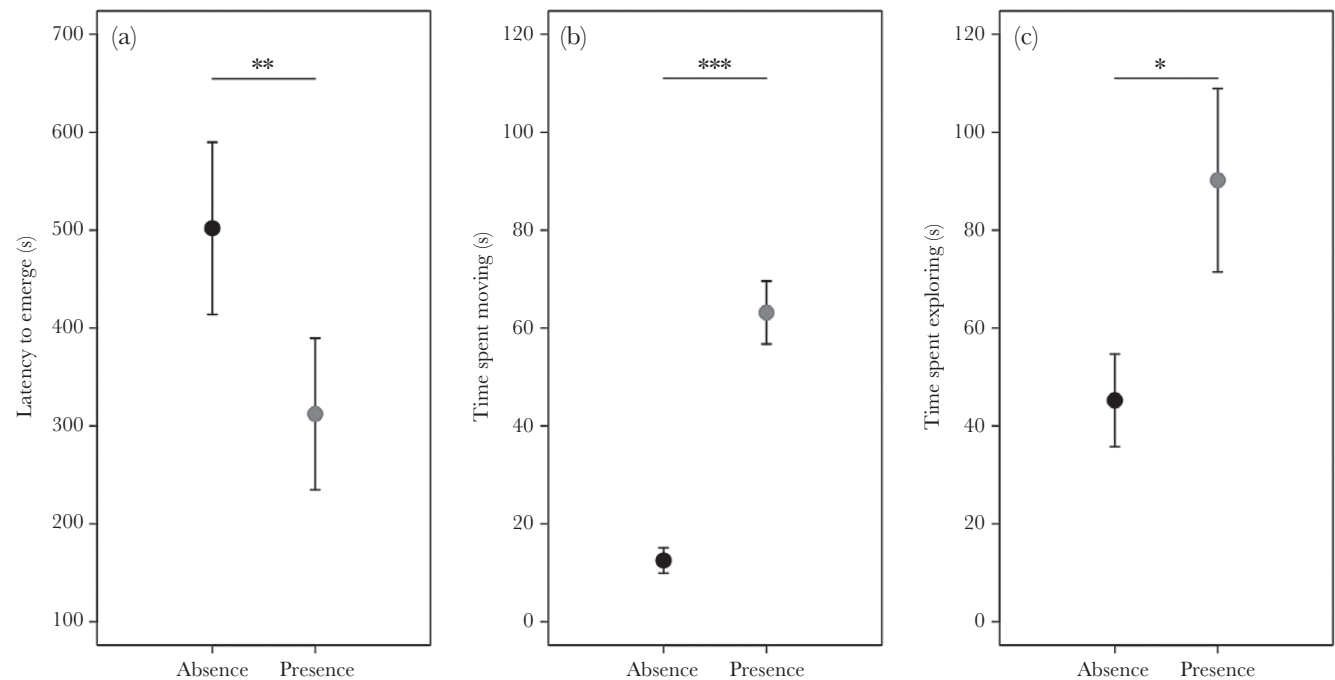


Figure 1
Mean values ($\pm$ standard errors) of behavioral traits expressed by offspring alone (black) or in the presence of their mother (gray). Behavioral traits include (a) boldness (i.e., latency to emerge), (b) activity (i.e., time spent moving), and (c) exploration (i.e., time spent exploring).

raised with their parent compared with those raised alone. First, maternal presence may provide a protected environment (see above) whereby offspring are able to allocate more resources into developing neurological structures possibly linked to learning later in life (Baarendse et al. 2013). Alternatively, studies on birds and mammals show that developing in social isolation can negatively affect cognitive development due to heightened levels of stress (e.g., Pryce et al. (2002) and Perez et al. (2012)). Overall, this corresponds with findings from obligate social species, where removing parental care during critical periods in offspring development can result in cognitive impairment (Pryce et al. 2002; Cacioppo and Hawkley 2009; Baarendse et al. 2013). Given that the majority of reptiles are highly precocial upon birth or hatching, few studies have examined how the early social environment affects cognitive development (Doody et al. 2013). A previous study on the effect of social isolation in reptiles found limited support for the suggestion that social isolation induces cognitive impairment (Riley et al. 2017). However, that study utilized same-aged conspecifics in the social treatment rather than the parents, potentially providing an explanation for the conflicting results between our study and theirs. As performance in learning tasks has been previously linked to increased survival and reproduction in reptiles (Dayananda and Webb 2017), our results suggest that there may be potential benefits of associating with your parents in terms of the development of key functional traits.

Given the clear differences in behavioral traits between treatments, we predicted that the presence of a mother would provide a number of fitness benefits to offspring. Indeed, previous work on this system using large-scale experimental studies has shown that offspring who remain in the maternal home range have increased growth rates compared with offspring who disperse (Botterill-James et al. 2016; Halliwell et al. 2017b). The differences in boldness, activity, exploration, and learning that we see in the presence of the mother provide a behavioral mechanism that could explain these results, specifically, that offspring can take advantage of greater access to foraging opportunities when in the presence of a parent. Interestingly, however, we found the opposite effect in this study; offspring housed with their mother grew significantly less than offspring who were held by themselves. One explanation for these results is the scale at which these respective experiments were carried out and the extent to which this influenced parent–offspring conflict. Indeed, the enclosures used in this experiment were small and had limited free-standing food, potentially creating greater competition for resources between parents and offspring than those carried out previously (which were carried out in large semi natural enclosures with abundant natural food resources). The possibility of local parent–offspring competition counteracting the benefits of facultative parental care has been relatively poorly studied, particularly in systems where offspring are not reliant on parental resources (Kramer et al. 2017; Kramer and Meunier 2018). However, there is some precedence for these results. For example, in the facultatively social European earwig (*Forficula auricularia*), maternal presence under food restriction triggered local competition between mothers and their offspring, resulting in substantial fitness costs to offspring (Kramer et al. 2017; see also Gardner and Smitseth 2011; Thesing et al. 2015). Combined, these results suggest that, even in systems where the costs of parental care are thought to be small (Botterill-James et al. 2016), there may be subtle changes in the cost and/or benefits afforded to individuals that depend on the environmental context. Ultimately, these changes can influence the extent to which

family life stabilizes. Exploring how these cost–benefit dynamics change under different environmental conditions will be a crucial next step in understanding the context dependent nature of social evolution.

A final caveat to our study is that our experimental design does not allow us to disentangle the potential effects of the presence of a mother versus the social environment per se. For example, offspring may exhibit plastic expression of behavior in the same direction as observed here had they been reared in the presence of adults that were not their parent (e.g., general social facilitation effects; Webster and Ward 2011). However, we argue that this distinction is irrelevant with respect to understanding the evolution of parental care. What is important is that the shift in social environment, related to the presence/absence of the mother, moderated the offspring's behavior and development in ways that may provide potential benefits to the offspring under certain environmental conditions. Given that parent–offspring associations are the most common form of social association that an offspring experiences during early life in this system (there is little evidence that offspring associate with nonparents; While et al. 2009a; Halliwell et al. 2017a), these positive effects of social associations are most likely to be borne out under the social circumstance we manipulated and thus represent a biologically relevant effect on offspring behavior that may feedback to stabilize this association between parents and offspring itself. Having said that, the extent to which these effects differ depending on whether the social environment contained related (e.g., the offspring's mother) versus an unrelated individual would be an extremely interesting avenue for future research and provide insights into the role that kin structure plays in mediating the emergence and maintenance of simple parent–offspring associations (see similar arguments in Ratcliff and Travisano (2014)).

In summary, our study shows that maternal presence in a facultatively social lizard can have a number of significant implications for the development and expression of key behavioral traits generally associated with increased fitness (Stamps 2007). Combined these results suggest that even relatively simple forms of parent–offspring associations can have impacts on offspring traits that may help refine and stabilize the expression of those associations, potentially setting the stage for the elaboration of both parent and offspring behaviors. However, our results also suggest that the extent to which this occurs may be dependent on whether the benefits of enhanced behavioral expression under parental presence outweigh any costs associated with enhanced parent–offspring competition (see also Thesing et al. (2015), Kramer et al. (2017), and Kramer and Meunier 2018). More generally, our findings highlight the importance of using simple facultative social systems for understanding the mechanisms and consequences of early social life (e.g., Falk et al. (2014), Kramer et al. (2017), Whiting and While (2017), and Kramer and Meunier 2018). An exciting avenue for future studies would be to examine whether the heightened levels of behavioral patterns found in this study affect reproduction and survival between philopatric and dispersing offspring. Doing so in a system where parental care is not obligatory and where we see a natural dichotomy between offspring receiving parental care and those that disperse provides us with a greater understanding of the simple steps in the early evolution of complex family living.

SUPPLEMENTARY MATERIAL

Supplementary data are available at *Behavioral Ecology* online.

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